

Comparing All Seven Parent Functions - Practice

Algebra II Honors | Unit 1 | Lesson 8 Practice | TEK A2.2.A

Objective: I can classify functions as even, odd, or neither by algebraic test, and find relative max/min on an interval.

Section 1 — Recall

1. Classify $\sqrt{x}$ as even, odd, or neither, and explain why it is "neither" for a different reason than 2^x is neither.

2. Is $f(x)$, x to the 4th power, even, odd, or neither? Show the algebra.

$f(-x) =$ _____

Section 2 — Apply

3. Using an algebraic test (not the graph), classify $g(x) = 1/x^2$ as even, odd, or neither.

$g(-x) =$ _____

4. $f(x) = x^3$ on $[-1, 3]$: give the minimum and maximum, and identify at which point(s) they occur.

min: _____ max: _____

Section 3 — Challenge

5. Two different parent functions from the seven are both "neither" (not even, not odd). Name them, and show the algebraic test for one of them.

6. Is it possible for an odd function to have a minimum value that is a positive number (like 3)? Reason using $f(-x) = -f(x)$ rather than picking a specific equation.

Section 4 — Explain It

7. A classmate claims that because x^3 and $\sqrt[3]{x}$ are both odd with no overall max or min, ANY two functions sharing those two properties must be reflections of each other across $y = x$. Is the classmate correct? Use $1/x$ as a counterexample or supporting case.
